

Lecture 3: Smoothed Analysis of Simplex via Circuits

Algorithms & Geometry of Linear Programming

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Worst-case instances are brittle

Klee–Minty (Lecture 2): **exponentially many** pivots — but brittle: small perturbations destroy the long path.

The smoothed model [Spielman–Teng '04]

Adversary: c and $(\bar{A}, \bar{b})$, rows of norm ≤ 1 ; nature: $\hat{A}, \hat{b}$ with i.i.d. $N(0, \sigma^2)$ entries.

$$\max c^T x \text{ s.t. } (\bar{A} + \hat{A})x \leq \bar{b} + \hat{b},$$

Bound the **expected** running time, worst case over the adversary.

Theorem (Spielman–Teng '04)

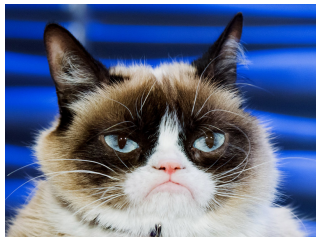
The shadow vertex simplex method solves smoothed LPs with expected $\text{poly}(n, m, 1/\sigma)$ pivots.

Sharper bounds: [Deshpande–Spielman '05], [Vershynin '09], [D–Huiberts '18], [Huiberts–Lee–Zhang '23].

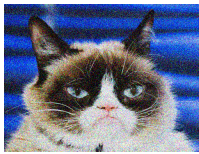
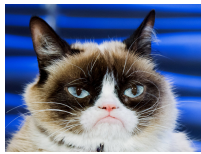
Random is not typical



Random is not typical



Smoothed complexity: [Spielman–Teng '04]



worst case, $\sigma = 0$

smoothed analysis, σ variable

Today: two complete smoothed analyses

Two complete analyses, perturbing only the **objective**:

1. **SSP** for minimum-cost flow
2. general LPs via **circuit imbalance**: constraint insertion

Common thread: **circuits** — simplex edge directions = flow cycles/paths.

Preliminaries

Smoothed Analysis of the Successive Shortest Path Algorithm

Circuit Decompositions & The Circuit Imbalance Measure

Smoothed Analysis via Circuit Imbalances

Outline

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The toolkit from Lecture 2

Standard form, $\mathbf{A} \in \mathbb{R}^{m \times n}$ of full row rank $m \leq n$:

$$\min c^T x \text{ s.t. } \mathbf{A}x = b, \quad x \geq \mathbf{0}.$$

- **basis** $B \subseteq [n]$, $|B| = m$, $\mathbf{A}_B$ invertible; **BFS**: $x_B = \mathbf{A}_B^{-1}b \geq \mathbf{0}$, $x_N = \mathbf{0}$ — a **vertex**
- **pivot directions** at basis B , entering $i \notin B$:

$$w_i^{B,i} = 1, \quad w_B^{B,i} = -\mathbf{A}_B^{-1}\mathbf{A}_i, \quad w_j^{B,i} = 0 \text{ otherwise}$$

— the **edges** at the vertex

- **variable bounds**: pair $x_j + \bar{x}_j = u_j$ — kernel notions carry over

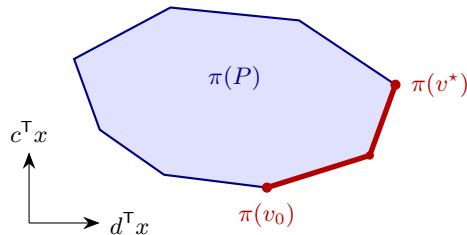
The shadow vertex rule (recap)

Shadow path for (c, d) : start objective c (minimizer v_0), **target** $-d$.

- maintain a vertex minimizing

$$c_\lambda := (1 - \lambda) c - \lambda d, \quad \lambda : 0 \rightarrow 1$$

- visited = **lower boundary** of the shadow $\pi(P)$, $\pi(x) = (d^\top x, c^\top x)$, walked rightward
- locality**: visited $\iff$ minimizes *some* c_λ — history-free



v_0 minimizes c , v^* maximizes d

The local step: the tangent cone program

$T_P(v) = \{y : v + \varepsilon y \in P \text{ for small } \varepsilon > 0\}$; standard form:
 $\{y \in \ker(\mathbf{A}) : y_j \geq 0 \ \forall j : v_j = 0\}$.

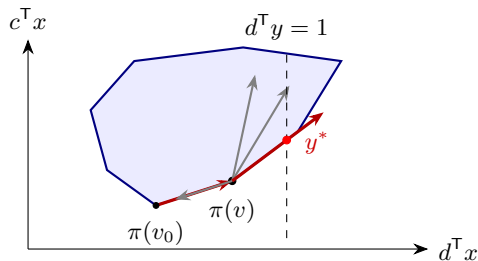
Lemma (Tangent cone program)

If v on the shadow path minimizes c_λ exactly for $\lambda \in [\lambda^-, \lambda^+]$, the edge followed out of v solves

$$\min c^\top y \text{ s.t. } y \in T_P(v), \quad d^\top y = 1,$$

with optimal value (slope) $\frac{\lambda^+}{1 - \lambda^+}$.

slopes **strictly increase** along the path



lowest cone direction of unit d -progress

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Minimum-cost maximum flow

This section: $G = (V, E)$ directed, $n := |V|$, $m := |E|$, source s , sink t , capacities $u > \mathbf{0}$, costs $c \in (0, 1)^E$.

- feasible flows:

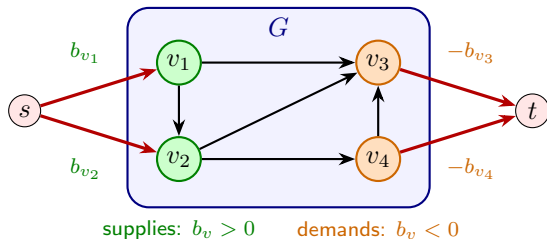
$$P := \left\{ f \in \mathbb{R}^E : \sum_{ij \in E} f_{ij} - \sum_{ji \in E} f_{ji} = 0 \quad \forall i \neq s, t, \quad \mathbf{0} \leq f \leq u \right\}$$

- $\text{val}(f) :=$ net flow leaving s
- incidence matrix $\mathbf{A}_G$: $(\mathbf{A}_G f)_v =$ net outflow at v
- goal:** min-cost flow among maximum-value flows

Reduction from minimum-cost flow

Min-cost b -flow, for demands $b \in \mathbb{R}^V$ with $\sum_v b_v = 0$:

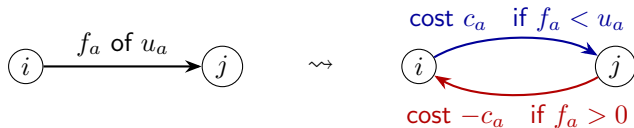
$$\min c^T f \text{ s.t. } \mathbf{A}_G f = b, \quad \mathbf{0} \leq f \leq u.$$



- auxiliary arcs (red): cost 0, capacity $|b_v|$
- max flows saturate all auxiliary arcs $\iff$ the instance is feasible

The residual network and the SSP algorithm

For $f \in P$, the *residual network* $N[f]$ keeps, for each arc $a = ij$:



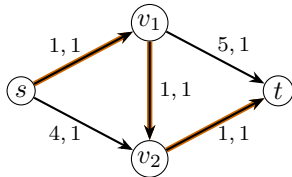
Successive shortest paths (SSP)

Start with $f = \mathbf{0}$. While $N[f]$ contains an s - t path: augment f as much as possible along a **minimum-cost** s - t path of $N[f]$. Return f .

Assume arising min-cost paths unique (probability 1 after perturbation).

SSP in action

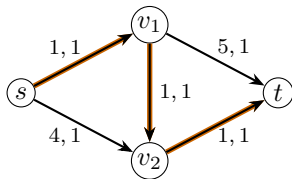
iteration 1 ($f = 0$)



arcs: cost, capacity
cheapest path: cost 3

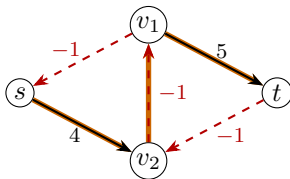
SSP in action

iteration 1 ($f = \mathbf{0}$)



arcs: cost, capacity
cheapest path: cost **3**

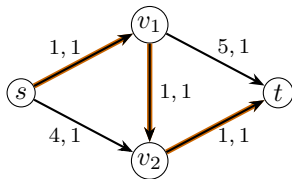
iteration 2: $N[f_1]$



dashed: **reverse** arcs
cheapest path: $4 - 1 + 5 = 8$

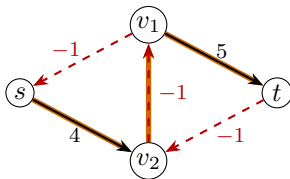
SSP in action

iteration 1 ($f = 0$)



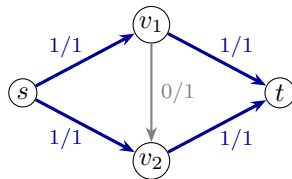
arcs: cost, capacity
cheapest path: cost 3

iteration 2: $N[f_1]$



dashed: reverse arcs
cheapest path: $4 - 1 + 5 = 8$

final flow (f_a/u_a)



no $s-t$ residual path:
min-cost max flow

iteration 2 **undoes** the flow on $v_1 v_2$; slopes increase: $3 < 8$.

SSP walks the cost curve

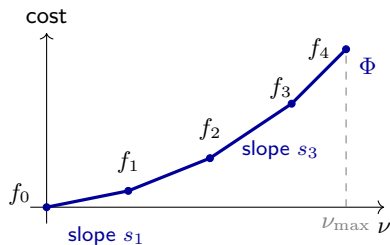
The *cost curve* $\Phi(\nu) := \min\{c^\top f : f \in P, \text{val}(f) = \nu\}$ is convex, piecewise linear on $[0, \nu_{\max}]$.

Theorem

The SSP iterates $f_0 = \mathbf{0}, f_1, \dots, f_k$ are exactly the breakpoints of Φ , in order, and the augmenting path costs s_i are its slopes:

$$0 < s_1 < \dots < s_k < n - 1.$$

- each f_i : min-cost of its value; last: min-cost max flow
- slopes: simple paths — $\leq n - 1$ arcs of cost in $(-1, 1)$



SSP implements the shadow vertex method

The shadow path for (c, val) walks the breakpoints of Φ — **already justified**. Remains: SSP performs its pivots.

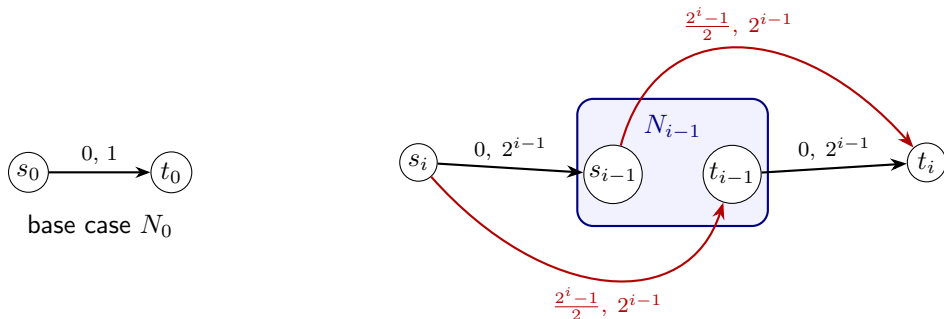
- the tangent cone program at a vertex f on the curve:

$$\begin{aligned} \min \quad & c^\top y \text{ s.t. } (\mathbf{A}_G y)_v = 0 \quad \forall v \neq s, t, \quad \text{val}(y) = 1, \\ & y_a \geq 0 \text{ if } f_a = 0, \quad y_a \leq 0 \text{ if } f_a = u_a \end{aligned}$$

- sign constraints = **residual arcs**: $N[f]$ is the tangent cone, combinatorially
- optimum = cheapest s - t path: one **shortest path** per pivot (exercise)
- [Zadeh '73; Disser-Skutella '19]: **exponentially many** iterations possible (next slide)

Exponentially many iterations [Disser–Skutella '19]

Recursive counting network N_i (arc labels: cost, capacity; their v_i 's set to 0):



recursive step: N_i from N_{i-1} , $i = 1, \dots, n$

- SSP on N_n : 2^n iterations, one unit of flow each
- the shortest s_n-t_n path at iteration j has cost exactly j , $j = 0, 1, \dots, 2^n - 1$

The perturbation model, and the theorem

Perturbation model

Each cost c_a , $a \in E$, is an independent continuous random variable on $(0, 1)$ with density at most $\phi \geq 1$. *Interval lemma*: $\Pr[c_a \in [\alpha, \beta]] \leq \phi (\beta - \alpha)$.

- example: uniform on an adversarial interval of length $1/\phi$
- $\phi \rightarrow \infty$: worst case — ϕ = adversary's **precision**

Theorem (Smoothed SSP, Brunsch et al. '15)

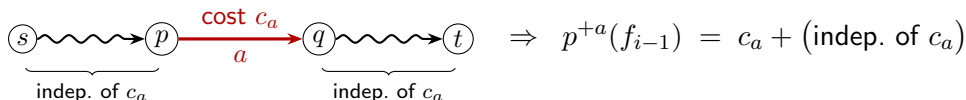
For any network with n nodes and m arcs, and random costs as above, the expected number of SSP iterations is $O(m n \phi)$.

Proof plan, and the charging step

Challenge: randomness not exhausted by the history. **Locality:** being on Φ is a property of the data.

- **arc sides** ($2m$): $ij \in E \leftrightarrow f_{ij} \geq 0, \quad ji \in \overleftarrow{E} \leftrightarrow f_{ij} \leq u_{ij} \quad (c_{ji} := -c_{ij})$
- **charging:** iteration i releases a tight bound — a side a of its path; charge i to a
- charged to $a \Rightarrow$ *the min-cost path uses a :*

$$p(f_{i-1}) = p^{+a}(f_{i-1}) = s_i$$



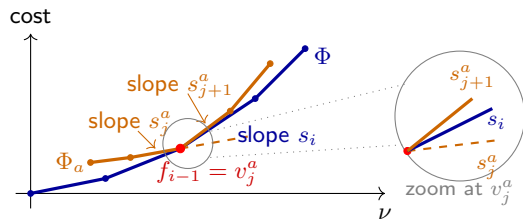
(p : min path cost in $N[f]$; p^{+a} : min over paths using a .) Plan: expected charge per side $O(n\phi)$, from c_a alone.

The sandwich step

Facet $P_a := \{f \in P : f_a \text{ at its bound}\}$: curve Φ_a , breakpoints v_j^a , slopes $s_j^a =$ **a -avoiding** path costs.

- charged to $a \Rightarrow f_{i-1}$ is a **breakpoint** v_j^a of Φ_a
- $\Phi \leq \Phi_a$, touching at f_{i-1} ; clip at 0 ($s_i > 0$):

$$\max(s_j^a, 0) \leq \underbrace{p^{+a}(v_j^a)}_{=s_i} \leq s_{j+1}^a$$



Distinct charges $\rightarrow$ distinct breakpoints:

$$\# \text{charged to } a \leq 2 + \sum_j \mathbf{1}[\max(s_j^a, 0) \leq p^{+a}(v_j^a) \leq s_{j+1}^a].$$

The probabilistic step

Let $e \in E$ underlie the side $a = pq$ ($c_a = \pm c_e$); condition on all costs but c_e .

- **independence:** Φ_a — breakpoints *and* slopes — does not depend on c_e
- **unit coefficient:** paths through a split as $(s-p) + (a) + (q-t)$:

$$p^{+a}(v_j^a) = c_a + r_j = \pm c_e + r_j, \quad r_j \text{ independent of } c_e$$

- event j : c_e in a **fixed interval** of length $\max(s_{j+1}^a, 0) - \max(s_j^a, 0)$
- **telescoping:** interval lengths sum to $\leq n - 1$

$$\mathbb{E}[\text{\#iterations}] \leq 2m (2 + (n - 1) \phi) = O(m n \phi). \quad \square$$

Where we are

Block 1:

- flow decomposition (proof still owed — next section)
- SSP = shadow vertex method walking Φ ; one shortest path per pivot
- $O(mn\phi)$ from **locality** + **bounded slope budget**

Block 2: general LPs — the **circuit toolkit**, then the algorithm.

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Smoothed Analysis via Circuit Imbalances

What did the analysis actually use?

Generic:

- charging pivots to entering coordinates
- independence of the facet curve from the charged cost
- the nested-curves sandwich

Flow-specific:

- augmenting structures with entries in $\{0, \pm 1\}$: charged cost enters with **unit coefficient**
- path costs $\leq n - 1$: slopes in a **bounded window**

Extension to $\min c^T x$, $Ax = b$, $x \geq 0$, smoothed c :

- unit coefficients $\rightarrow$ the **circuit imbalance measure** $\kappa(A)$
- bounded window $\rightarrow$ **algorithm design**: insert constraints *one at a time*

Circuits of a subspace

Definition (Circuit)

Let $W \subseteq \mathbb{R}^n$ be a linear subspace. A *circuit* of W is a support-minimal nonzero $g \in W$: no nonzero $h \in W$ has $\text{supp}(h) \subsetneq \text{supp}(g)$.

Circuits of a *matrix* $\mathbf{A}$: the circuits of $\ker(\mathbf{A})$ — the **minimal linear dependencies** among its columns.

- depends only on the subspace ($\ker(\mathbf{A})$ here; $\text{im}(\mathbf{A}^T)$ works too)
- **role 1**: circuits are the **pivot directions** of the simplex method (later this section)
- **role 2**: circuits are the elementary pieces into which every vector of W decomposes **without cancellation** (next slide)

For incidence matrices: circuits = *simple cycles* — two slides on.

Conformal circuit decomposition

Definition (Conformal vectors)

g conforms to w if $\text{supp}(g) \subseteq \text{supp}(w)$ and $g_j w_j \geq 0$ for all j (signs agree).

Lemma (Conformal decomposition)

Every $w \in W \setminus \{0\}$ is a sum $w = g^1 + \dots + g^p$ of $p \leq \dim(W)$ circuits of W conforming to w ; cancellation-free: $|w_j| = \sum_i |g_j^i|$.

Proof via the **pointed** cone

$$C_w := \{y \in W : y_j w_j \geq 0 \ \forall j \in \text{supp}(w), \quad y_j = 0 \ \forall j \notin \text{supp}(w)\} \ni w :$$

- **Minkowski:** $C_w = \text{cone}(R(C_w))$
- **extreme rays are circuits:** the tight constraints of a 1D face cut out $\{y \in W : \text{supp}(y) \subseteq \text{supp}(g)\} = \text{span}(g)$
- **Carathéodory:** keep independent generators — $p \leq \dim(W)$

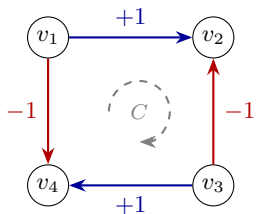
Circulations and their circuits

Specializing to the incidence matrix $\mathbf{A}_G$:

- $\ker(\mathbf{A}_G) = \text{circulations}$ (negative entries = flow against the arc)

Lemma

The circuits of $\mathbf{A}_G$ are exactly the nonzero multiples of the signed indicators χ^C of *simple cycles* C .



χ^C : $+1$ on forward arcs,
 -1 on backward arcs

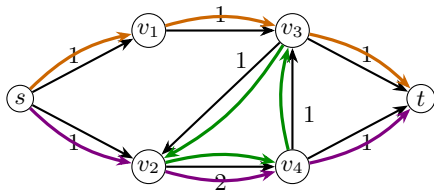
Flow decomposition

Theorem (Flow decomposition)

Every $f \in P$ with $\text{val}(f) = \nu \geq 0$ decomposes as

$$f = \sum_i t_i \chi^{Q_i} + \sum_j t'_j \chi^{C_j}, \quad t, t' > 0, \quad \sum_i t_i = \nu,$$

with $\leq m + 1$ simple **directed** s - t paths Q_i and cycles C_j : all arcs in $\text{supp}(f)$, traversed forwards. More generally, every circulation is a positive sum of $\leq m$ simple cycles **conforming** to it (forwards where $w_a > 0$, backwards where $w_a < 0$).



$$f = \chi^{Q_1} + \chi^{Q_2} + \chi^{C_1}: \quad \text{two directed } s\text{-}t \text{ paths and one directed cycle, } \text{val}(f) = 2$$

The circuit imbalance measure

Definition (Circuit imbalance)

For a linear subspace $W \subseteq \mathbb{R}^n$,

$$\kappa(W) := \max \left\{ \frac{|g_j|}{|g_i|} : g \text{ a circuit of } W, i, j \in \text{supp}(g) \right\},$$

the largest ratio between two entries of a circuit ($\kappa := 1$ if $W = \{\mathbf{0}\}$);
 $\kappa(\mathbf{A}) := \kappa(\ker(\mathbf{A}))$.

- incidence matrices, and all totally unimodular $\mathbf{A}$: $\kappa = 1$
- integral $\mathbf{A}$: $\kappa \leq \Delta_{\max}(\mathbf{A})$
- invariant under row operations; *not* under column scaling

(next slide)

Circuits are the pivot directions

Proposition

Every pivot direction $w^{B,i}$ is a circuit, and every circuit of $\mathbf{A}$ is a scalar multiple of some $w^{B,i}$. Consequently

$$\kappa(\mathbf{A}) = \max \left\{ |(\mathbf{A}_B^{-1} \mathbf{A}_i)_j| : B \text{ basis}, i \notin B, j \in B \right\}.$$

- $w^{B,i}$ support-minimal: smaller support $\Rightarrow$ supported on $B \Rightarrow$ zero
- circuit g : extend $\text{supp}(g) \setminus \{i\}$ to a basis B ; then $g/g_i = w^{B,i}$
- used constantly: $|\text{supp}(g)| \leq m+1, \quad \|g\|_1 \leq (m+1) \kappa |g_\ell|$
- **Cramer**: $(\mathbf{A}_B^{-1} \mathbf{A}_i)_j = \pm \det(\mathbf{A}_{B-j+i}) / \det(\mathbf{A}_B)$: $\kappa = \text{max determinant ratio}$ of adjacent bases $\Rightarrow \kappa \leq \Delta_{\max}$ for integral $\mathbf{A}$

Stability of κ

Lemma (Stability)

$\kappa(\mathbf{A})$ does not increase under:

- (i) deleting a column;
- (ii) contracting a column i : row-reduce it to some e_j , then delete row j and column i ;
- (iii) duplicating a column up to sign: append $\pm \mathbf{A}_i$ as a new variable y .

- (i): circuits of the smaller matrix are circuits of $\mathbf{A}$ avoiding column i
- (ii): on kernels, contraction *projects coordinate i away*; a minimal-support lift of a circuit is a circuit of $\mathbf{A}$
- (iii): the only new circuit is the pair (x_i, y) , of ratio 1

Exactly the operations relating the phase systems of the coming algorithm to $\mathbf{A}$.

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Inserting one constraint at a time

Assumptions: c_j independent, density $\leq \phi$ on $(0, 1]$; $\mathbf{A}_{[m]} = \mathbf{I}_m$ (row ops + permutation); truncated systems non-degenerate.

For $k \in [m]$, the *phase- k trade-off curve* (convex, piecewise linear):

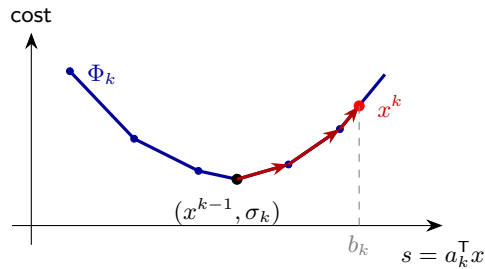
$$\Phi_k(s) := \min \left\{ c^\top x : \mathbf{A}_{[k-1],[n]} x = b_{[k-1]}, \quad a_k^\top x = s, \quad x \geq \mathbf{0} \right\}, \quad s \in I_k.$$

Constraint-insertion shadow simplex

Set $x^0 := \mathbf{0}$. For $k = 1, \dots, m$: starting from $(x^{k-1}, \sigma_k := a_k^\top x^{k-1})$, walk Φ_k monotonically from $s = \sigma_k$ to $s = b_k$; if $b_k \notin I_k$, stop and report *infeasible*. Return x^m .

The phases chain together

- **idle variables rest:** on Φ_k , $x_j = 0$ for $j > k$
- **chaining:** $(x^{k-1}, \sigma_k) =$ **global minimum** of Φ_k (s is free)
- each phase starts at the bottom, climbs to level b_k



Complete solver: $b_k \notin I_k \Rightarrow$ infeasible; no Phase 0 needed.

The phase systems inherit κ

In variables (x, s) , phase k has constraint matrix

$$\mathbf{M}_k := \left[\mathbf{A}_{[k],[n]} \mid -e_k \right] \quad \left(\mathbf{A}_{[k-1]}x = b_{[k-1]}, \quad a_k^\top x - s = 0 \right).$$

- **design point:** $-e_k$ duplicates a present column; the interpolation functional is the *coordinate* s
- “ $d^\top w$ ” is an *entry of the circuit*: no cancellations

Lemma

$\kappa(\mathbf{M}_k) \leq \kappa(\mathbf{A})$ for every k , and every circuit of $\mathbf{M}_k$ has support of size at most $k + 1$.

The smoothed guarantee

Theorem (Smoothed constraint insertion)

Phase k performs at most $n(2 + (k + 1)\kappa^2\phi)$ pivots in expectation; over all m phases,

$$\mathbb{E}[\#\text{pivots}] \leq 2nm + 2nm^2\kappa^2\phi = O(nm^2\kappa^2\phi).$$

The SSP proof applies *word for word* — the dictionary:

SSP	phase k of constraint insertion
iteration	pivot along the curve Φ_k
arc-side a (at most $2m$)	entering coordinate $a \in [n]$
facet curve Φ_a (paths avoiding a)	curve $\Phi_{k,a}$ (column a deleted)
clipped slope window $n - 1$	clipped slope window $(k + 1)\kappa$
$p^{+a} = \pm c_a + r_j$ (unit coefficient)	slope derivative $w_a/w_s \in [1/\kappa, \kappa]$

The dictionary, in detail

- **slopes:** each edge is a circuit (w, w_s) of $\mathbf{M}_k$:

$$|c^\top w / w_s| \leq \|c\|_\infty \|w\|_1 / |w_s| \leq (k+1)\kappa$$

- **charging:** to the entering coordinate a (bound $x_a = 0$ released)
- **sandwich:** $\Phi_{k,a}$ (column a deleted) involves neither x_a nor c_a
- **sweep:** entering- a slope affine in c_a , derivative $w_a/w_s \in [1/\kappa, \kappa] \Rightarrow$ one interval of length $\leq \kappa \cdot \text{gap}$

Telescope over the clipped window $(k+1)\kappa$: expected charge $\leq 2 + (k+1)\kappa^2\phi$ per coordinate. □

Discussion

- $\kappa = 1$: bound $O(nm^2\phi)$ — a factor m above SSP; amortizing the phases is **open**
- why a fresh coordinate: a fixed d can pair *degenerately* with circuits ($d^T w \approx 0$); a coordinate cannot
- alternative: randomize the interpolation objectives — works for worst-case c [Bruns–Röglin '13; Dadush–Hähnle '16]

Summary and what comes next

Today:

- *circuits*: pivot directions = flow cycles/paths; the measure κ
- *conformal decomposition* (Minkowski + Carathéodory) $\Rightarrow$ flow decomposition
- SSP: $O(mn\phi)$; constraint insertion: $O(nm^2\kappa^2\phi)$ — *locality + bounded slope budget*

Next lecture: *interior point methods* — unconditional guarantees; circuits reappear.

Exercises: SSP implements the shadow vertex method; source and sink arcs; flow decomposition from conformal decomposition; self-duality of κ .